Boundary maxwell module

We are solving:

$$\begin{cases} \nabla\times\left(\dfrac{1}{\mu}\nabla\times E\right)-\omega^2\varepsilon E = J \quad \text{in}\,\Omega \\ \nabla\left(\varepsilon E\right)=0 \\ \varepsilon\in\mathbb{C} \\ \varepsilon,\mu = f(x,y,z) \end{cases}$$

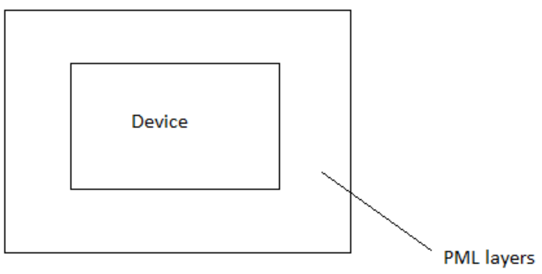
On each boundary is considered one of the following conditions:

$$n\times\left(\frac{1}{\mu}\nabla\times E\right)=0 \qquad \text{Electric wall}$$
$$n\times E = 0 \quad on\ \Gamma_1 \qquad \text{Magnetic wall}$$

Using finite element equation is transformed to system of linear equations.

PML layers:

PML layers are used to solve discontinuous problems. They should surround device:



External shape of the PML boundary should be "good", i.e. in 2D if PML surrounds all the device it should be rectangular. PML layers absorb light but give no reflection. Thickness of PML can be around half of the wavelength.

Source condition

Source of light. J in equation. Only plane waves are considered. Line source in 2D, surface source in 3D.

Input file example

Module boundarymaxwell {

approxOrder **= 2**	Default 0. Order of approximation. The higher order is the higher is the matrix dimension. So the error decreases and solving time increases.
extraOrder = 0	Default 0. Additional order for numerical integration. No sense to use it in current implementation -)

W	Frequency of light in eV.
inplane	Default 1. Used in 2D. If problem is solved for inplane electric field or outplane.
Contact ewall {	
type = ElectricWall	Tangent component of electric field is zero.
}	
Contact mwall {	
type = MagneticWall	Tangent component of magnetic field is zero.
}	
Contact source {	
type = Source	Source of light.
direction=0	Direction of light. Default is y in 1D, z in 2D outplane. 0 for x, 1 for y, 2 for z.
power=1	Strength of light source. Make sense if several sources are considered. Power of 1 gives electric field of 1 in vacuum.
}	

}

Material properties:

[permittivity]

optical_mu	Permeability.
sPML	Characterize absorption in PML layers. Value can be set to 6.
optic_epsilon	Default 1. Dielectric constant.
optic_epsilon_imag	Default 0. Imaginary part of dielectric constant.

Instead you can define:

optic_epsilon00, optic_epsilon11, optic_epsilon22

optic_epsilon_imag00, optic_epsilon_imag11, optic_epsilon_imag22

Output:

Efield	Electric field.
Epsilon	epsilon_x_x
Epsilon_imag	epsilon_imag_x_x
Mu	mu
Efield_real, Efield_imag	Contains information about phase. Obviously $$E_{field} = \sqrt{E_{field\_imag}{}^2 + E_{field\_real}{}^2}$$